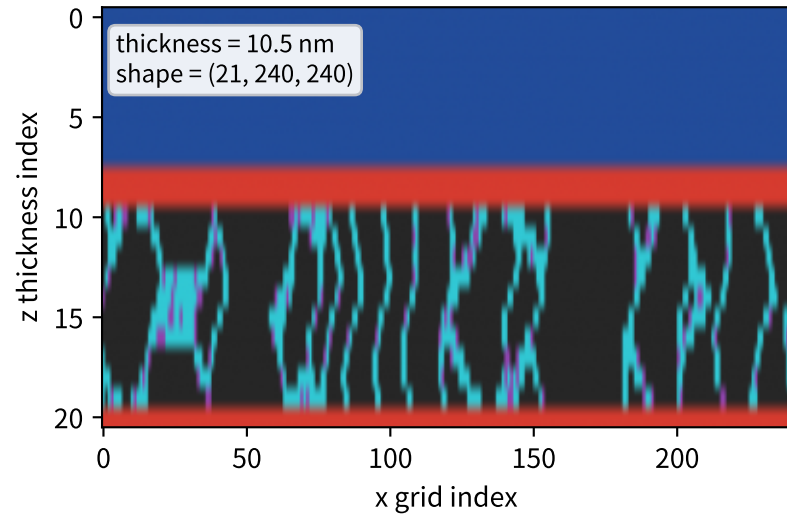


3D Al₂O₃/ZnO bilayer kMC: readable transport analysis

Rare Al₂O₃ pinholes + interface lateral search + ZnO boundary/channel tortuosity make successful transmission uncommon.

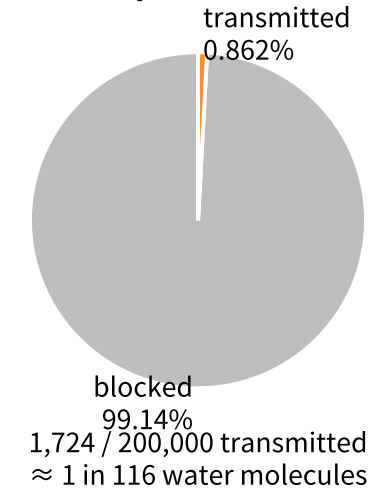
Middle 3D slice: grid[z, x, y_{mid}=120]



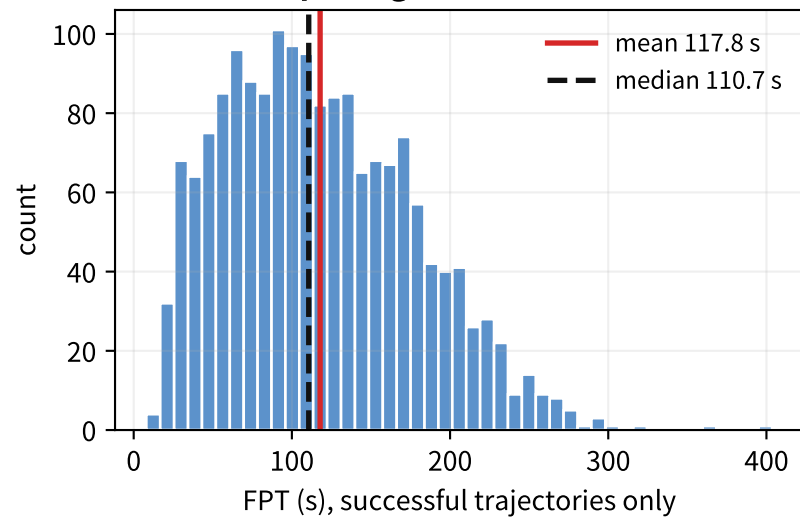
Cell types

- Al₂O₃ dense matrix
- Al₂O₃ pinhole
- Al₂O₃/ZnO interface
- ZnO grain/bulk
- ZnO GB/channel
- blocked/dead end

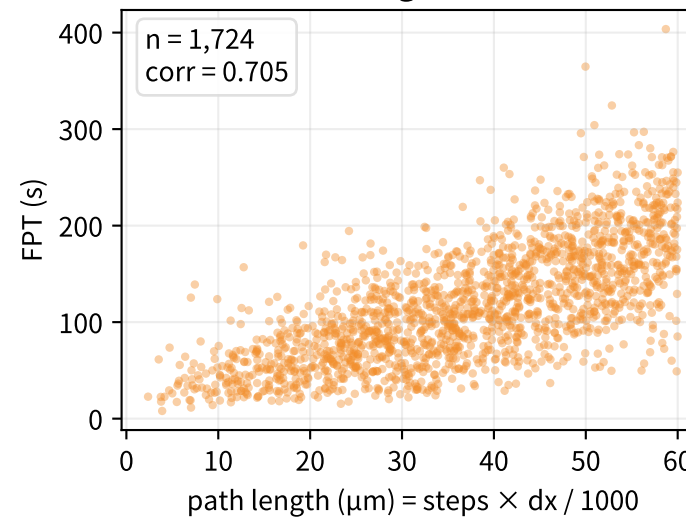
Transport outcome



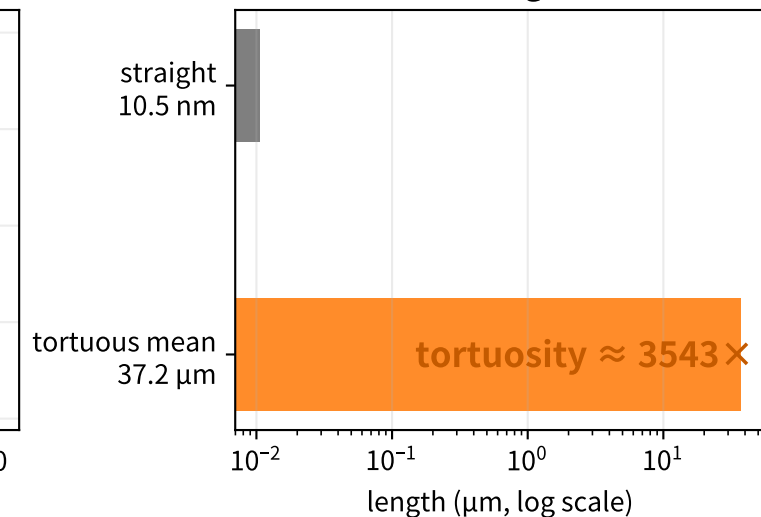
First-passage time distribution



Path length vs FPT



Path elongation



P_{trans}

mean FPT

FPT p₁₀–p₉₀

mean path

path p₁₀–p₉₀

pinhole area

0.862% (1,724/200,000)

117.8 s

44.2–201.2 s

37.2 μm

17.2–55.8 μm

0.0677%

Mechanism: Most water molecules do not enter because Al₂O₃ pinholes are rare. A successful molecule must find a pinhole, search laterally at the Al₂O₃/ZnO interface, then follow a long boundary/channel path while avoiding blocked dead ends. Therefore geometric thickness is only 10.5 nm, but successful paths average ~37.2 μm.